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Modeling the Impact of Environmental Factors on Ecosystem Stress Using Multiple Regression

Case Study 5 — Multiple Regression in Biology & Analytics
STAT 482 — Capstone in Analytics for Business

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Table of Contents

1. Introduction

Wildfires represent one of the most significant ecological disturbances affecting terrestrial ecosystems worldwide. Beyond the immediate destruction of vegetation and fauna, large fire events alter soil chemistry, watershed dynamics, and long-term land productivity — making them a critical proxy for ecosystem stress. As climate change intensifies drought conditions and raises ambient temperatures across Mediterranean and semi-arid biomes, understanding the environmental drivers of fire-induced land damage has become both a scientific imperative and a practical necessity for land managers, agricultural planners, and disaster-risk policymakers.

This report presents a comprehensive multiple linear regression analysis of the UCI Forest Fires dataset, which records 517 fire events from Montesinho Natural Park in northeast Portugal. The dependent variable — burned area (log-transformed as log(area + 1)) — serves as a proxy for ecosystem impact. Eight continuous environmental predictors are examined: temperature (temp), relative humidity (RH), wind speed (wind), rainfall (rain), Fine Fuel Moisture Code (FFMC), Duff Moisture Code (DMC), Drought Code (DC), and Initial Spread Index (ISI).

The analysis proceeds through seven structured stages: data understanding and cleaning, exploratory data analysis (EDA), model specification, multicollinearity diagnostics, model assumption testing, stepwise model selection, and out-of-sample validation. The ultimate aim is to identify the environmental conditions most predictive of burned area and to translate those statistical findings into actionable biological and managerial insights.

2. Data Description and Cleaning

2.1 Dataset Overview

The Forest Fires dataset contains 517 observations and 13 variables. After loading the data in R and running dim(fires) and str(fires), the structure confirms 517 rows and 13 columns including spatial coordinates (X, Y), temporal identifiers (month, day), four Fire Weather Index (FWI) system components (FFMC, DMC, DC, ISI), four meteorological measurements (temp, RH, wind, rain), and the response variable area. A call to colSums(is.na(fires)) confirms zero missing values across all 13 variables, meaning no imputation was required.

The categorical variables month and day were converted to ordered factors in R to reflect their natural chronological ordering. Summary statistics for the key numeric variables are presented in Table 1 below.

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
FFMC	18.70	90.20	91.60	90.64	92.90	96.20
DMC	1.10	68.60	108.30	110.90	142.40	291.30
DC	7.90	437.70	664.20	547.90	713.90	860.60
ISI	0.00	6.50	8.40	9.02	10.80	56.10
temp (°C)	2.20	15.50	19.30	18.89	22.80	33.30
RH (%)	15.00	33.00	42.00	44.29	53.00	100.00
wind (km/h)	0.40	2.70	4.00	4.02	4.90	9.40
rain (mm)	0.00	0.00	0.00	0.02	0.00	6.40
area (ha)	0.00	0.00	0.52	12.85	6.57	1090.84

Table 1. Descriptive statistics for all numeric predictors and the response variable (N = 517).

2.2 Outlier Detection and Response Transformation

Boxplots of all numeric variables (Figure 1 in the R output) reveal that the burned area variable is severely right-skewed. The mean area of 12.85 ha is far above the median of 0.52 ha, and the maximum recorded fire reached 1,090.84 ha — a clear extreme outlier. A similar, though less severe, right skew is visible in DMC and rain. The X and Y spatial grid coordinates show minimal dispersion, confirming they are bounded categorical-like integers.

To address the non-normality of the response, a standard log-plus-one transformation was applied: log_area = log(area + 1). The histograms generated in R (Figure 2) confirm that the log-transformed distribution is substantially more symmetric and approaches a bell shape, making it appropriate as the dependent variable in a linear regression framework. After transformation, the working range of log_area spans approximately 0 to 7.

3. Exploratory Data Analysis

3.1 Correlation Structure

A correlation matrix was computed among all numeric variables including log_area (Table 2). The most notable inter-predictor correlations involve the FWI system components: DMC and DC share a correlation of 0.68, while FFMC and ISI are correlated at 0.53. Temperature exhibits moderate positive correlations with DMC (0.47) and DC (0.50), and a strong negative relationship with relative humidity (−0.53). These patterns are ecologically sensible — hotter and drier conditions simultaneously raise both the moisture-deficit indices and reduce atmospheric humidity.

With respect to the response variable log_area, no single predictor exceeds a Pearson correlation of 0.07 in absolute magnitude. The strongest bivariate associations are area vs. log_area ($r = 0.52$, by construction) and DMC vs. log_area ($r = 0.07$). This already suggests that the predictive signal is weak and distributed rather than concentrated in one or two drivers — a finding that will be confirmed during model estimation.

	FFMC	DMC	DC	ISI	temp	RH	wind	rain	log_area
FFMC	1.00	0.38	0.33	0.53	0.43	-0.30	-0.03	0.06	0.05
DMC	0.38	1.00	0.68	0.31	0.47	0.07	-0.11	0.07	0.07
DC	0.33	0.68	1.00	0.23	0.50	-0.04	-0.20	0.04	0.07
ISI	0.53	0.31	0.23	1.00	0.39	-0.13	0.11	0.07	-0.01
temp	0.43	0.47	0.50	0.39	1.00	-0.53	-0.23	0.07	0.05
RH	-0.30	0.07	-0.04	-0.13	-0.53	1.00	0.07	0.10	-0.05
wind	-0.03	-0.11	-0.20	0.11	-0.23	0.07	1.00	0.06	0.07
rain	0.06	0.07	0.04	0.07	0.07	0.10	0.06	1.00	0.02
log_area	0.05	0.07	0.07	-0.01	0.05	-0.05	0.07	0.02	1.00

Table 2. Pearson correlation matrix for all numeric variables. Values rounded to two decimal places.

3.2 Scatterplots Against the Response

The faceted scatterplots of each predictor against log_area (Figure 3 in the R output) confirm the weak bivariate relationships indicated by the correlation matrix. Across all eight predictors, the fitted linear trend lines are nearly flat. Temperature shows a very slight positive trend, while relative humidity displays a marginally negative association. Rain observations cluster almost entirely at zero, reflecting the large proportion of fire events that occurred on dry days. These flat trends reinforce the conclusion that predicting burned area from individual environmental variables is inherently difficult — a phenomenon consistent with the existing wildfire science literature, which notes that burned area is also heavily influenced by landscape topology, vegetation density, and fire suppression response time, none of which are captured in this dataset.

4. Model Specification and Estimation

4.1 Full Regression Model

The primary model regresses log_area on all eight continuous predictors using ordinary least squares (OLS):

$$\log(area + 1) = \beta_0 + \beta_1 \cdot FFMC + \beta_2 \cdot DMC + \beta_3 \cdot DC + \beta_4 \cdot ISI + \beta_5 \cdot temp + \beta_6 \cdot RH + \beta_7 \cdot wind + \beta_8 \cdot rain + \varepsilon$$

The full model results are summarised in Table 3. The intercept is 0.222 ($p = 0.870$), and the F-statistic of 1.288 on 8 and 508 degrees of freedom yields a p-value of 0.247, indicating that the overall model is not statistically significant at the conventional 5% level. The Multiple R^2 is 0.020 and the Adjusted R^2 is only 0.004 — meaning the eight predictors collectively explain less than 0.5% of the variance in log-transformed burned area after adjusting for the number of predictors.

Among individual coefficients, wind speed is the only predictor that reaches statistical significance ($\beta = 0.076$, $t = 2.069$, $p = 0.039$). Its positive sign indicates that higher wind speeds are associated with larger burned areas on the log scale — consistent with the physical role of wind in accelerating fire spread and reducing fuel moisture through increased evaporation. No other predictor achieves $p < 0.05$ in the full model.

Predictor	Estimate	Std. Error	t-value	p-value	Significance
(Intercept)	0.2224	1.3604	0.163	0.870	
FFMC	0.0077	0.0145	0.532	0.595	
DMC	0.0012	0.0015	0.814	0.416	
DC	0.0003	0.0004	0.767	0.444	
ISI	-0.0239	0.0169	-1.415	0.158	
temp	0.0025	0.0173	0.143	0.887	
RH	-0.0052	0.0052	-0.997	0.319	
wind	0.0758	0.0366	2.069	0.039	*
rain	0.0965	0.2121	0.455	0.649	

Table 3. Full OLS regression results. Dependent variable: log(area + 1). N = 517. Residual SE = 1.395 on 508 df. Adjusted R^2 = 0.004. F(8,508) = 1.288, p = 0.247. (*) p < 0.05.

4.2 Extended Model with Temporal Controls

A second model (model_time) adds month and day as ordered categorical factors. This extended specification yields an Adjusted R^2 of 0.021 — a modest improvement over the base model. The F-statistic is 1.442 on 25 and 491 degrees of freedom ($p = 0.078$), marginally approaching significance. Within this model, the DMC coefficient becomes significant ($\beta = 0.004$, $t = 2.187$, $p = 0.029$), suggesting that after controlling for seasonal and day-of-week effects, the Duff Moisture Code does carry information about fire severity. Temperature approaches significance ($p = 0.095$), consistent with its expected biological role. The month^7 term (July polynomial component) yields a coefficient of 1.354 and $p = 0.054$, hinting at elevated fire risk in mid-summer — a pattern with clear practical implications for fire management resource allocation.

4.3 Interaction Model

A third model introduces a temp × RH interaction term to test whether the combined effect of high temperature and low humidity amplifies burned area beyond the additive contributions of each variable alone. The interaction coefficient is −0.00049 ($t = -0.763$, $p = 0.446$), which is not significant. The model's Adjusted R^2 (0.004) is lower than even the base full model, and the AIC (1824) is intermediate. These results suggest that a linear interaction between temperature and humidity does not provide meaningful additional predictive power in this dataset.

5. Multicollinearity Diagnostics

Variance Inflation Factors (VIF) were computed for the full model using the car package. The results are reported in Table 4. All VIF values fall below the threshold of 5, indicating that multicollinearity does not pose a serious concern for the stability of the regression coefficients in this model. The highest VIF belongs to temperature (2.66), followed by DC (2.08) and DMC (2.33) — reflecting the moderate inter-correlations among the FWI components and their shared dependence on temperature and moisture conditions noted during EDA. Wind speed and rainfall exhibit near-unity VIF values (1.14 and 1.04 respectively), confirming their statistical independence from the other predictors.

Predictor	VIF Value	Interpretation
FFMC	1.695	Acceptable (< 5)
DMC	2.331	Acceptable (< 5)
DC	2.078	Acceptable (< 5)
ISI	1.578	Acceptable (< 5)
temp	2.662	Acceptable (< 5)
RH	1.900	Acceptable (< 5)
wind	1.141	Acceptable (< 5)
rain	1.045	Acceptable (< 5)

Table 4. Variance Inflation Factors for the full regression model. VIF < 5 indicates acceptable multicollinearity.

These findings have an important implication: the weak predictive performance of the full model (Adjusted $R^2 \approx 0.004$) cannot be attributed to inflated standard errors caused by multicollinearity. Rather, it reflects a genuine lack of linear signal between the measured environmental conditions and fire extent — a conclusion that is consistent with the existing literature on forest fire prediction, which suggests that burned area is inherently difficult to model using only weather and fuel moisture conditions without spatial and topographic covariates.

6. Model Assumption Diagnostics

6.1 Residuals vs. Fitted Plot (Linearity and Homoscedasticity)

The Residuals vs. Fitted plot from the full model reveals several concerning patterns. The residuals are not randomly distributed around zero across the range of fitted values; instead, a fan-shaped spread is visible, with larger residuals at higher fitted values. This is indicative of heteroscedasticity — non-constant variance of the error term. Observations 239, 416, and 480 are flagged as potential outliers with large positive residuals, corresponding to fires with unusually large burned areas given their predictor values.

6.2 Breusch-Pagan Test

To formally test for heteroscedasticity, the Breusch-Pagan test was applied to the full model. The test yields BP = 11.27, df = 8, p-value = 0.187. Since $p > 0.05$, we fail to reject the null hypothesis of homoscedasticity at the 5% significance level. This result is reassuring: despite the visual suggestion of a fan pattern in the residuals, the formal statistical test does not detect significant heteroscedasticity. This may reflect the relatively modest power of the test or the effectiveness of the log transformation in stabilising variance. The finding means that standard OLS standard errors remain valid for inference purposes.

6.3 Q-Q Plot and Normality Test (Shapiro-Wilk)

The Normal Q-Q plot displays a pronounced departure from normality, particularly in the upper tail where many points deviate above the reference line. This is consistent with the presence of a small number of very large fire events that, even after log transformation, remain extreme relative to the bulk of the distribution. The Shapiro-Wilk test formalises this finding: $W = 0.854$, p-value $< 2.2 \times 10^{-16}$. The null hypothesis of normality is decisively rejected.

This non-normality of residuals has implications for hypothesis testing: p-values and confidence intervals derived from the t-distribution may be unreliable, particularly for tests involving extreme observations. However, for large samples ($N = 517$), the Central Limit Theorem provides some protection against non-normality for inference on the regression coefficients. Robust standard errors or bootstrap inference would be advisable for formal hypothesis testing in a production setting.

6.4 Scale-Location and Leverage

The Scale-Location plot confirms the heteroscedastic tendency noted visually in the Residuals vs. Fitted plot — the red smoothing line trends upward, indicating that the spread of standardised residuals increases with fitted values. The Residuals vs. Leverage plot identifies observation 500 as having the highest leverage (approximately 0.88), though it lies within Cook's distance contours, meaning it does not exert undue influence on the fitted coefficients. Observations 416 and 239 have high standardised residuals but relatively modest leverage, classifying them as outliers rather than influential points.

7. Model Selection

7.1 Comparison of Candidate Models

Three candidate models were evaluated using Adjusted R^2 and information criteria (AIC and BIC). The results, summarised in Table 5, show that the Time Controls model (which includes month and day factors) achieves the highest Adjusted R^2 (0.021) and is the only model whose F-test p-value approaches significance ($p = 0.078$). However, it pays a penalty in both AIC (1830) and BIC (1945) — substantially higher than the Full Model (AIC = 1823, BIC = 1865) — because adding 17 additional categorical parameters is not justified by the modest improvement in fit.

Model	Adj. R^2	AIC	BIC	F p-value
Full Model (8 predictors)	0.004	1823	1865	0.247
Time Controls (+month, day)	0.021	1830	1945	0.078
Interaction Model (temp × RH)	0.004	1824	1871	0.287
Stepwise Model (DMC + RH + wind)	0.008	1816	1837	0.061

Table 5. Model comparison by Adjusted R^2, AIC, and BIC. Dependent variable: log(area + 1).

7.2 Stepwise Regression

Bidirectional stepwise regression (stepAIC with direction = 'both') was applied to the full model. The procedure retained three predictors — DMC, RH, and wind — as the optimal subset, producing the following estimated equation:

$$\log(area + 1) = 0.913 + 0.00175 \cdot DMC - 0.00558 \cdot RH + 0.0624 \cdot wind$$

This parsimonious model achieves an AIC of 1815.52 and BIC of 1836.76 — the lowest of any candidate model, reflecting the best balance between fit and complexity under both criteria. The Adjusted R^2 is 0.008 and the overall F-statistic is 2.472 ($p = 0.061$), approaching significance at the 10% level. The three retained predictors have the following interpretations. The positive DMC coefficient (0.00175) indicates that drier soil conditions, as measured by the Duff Moisture Code, are associated with larger fires. The negative RH coefficient (−0.00558) reflects that lower relative humidity — typical of hot, dry weather — corresponds to more extensive burning. The positive wind coefficient (0.0624) confirms the key physical role of wind speed: for each 1 km/h increase in wind speed, log_area increases by approximately 0.062 units, holding DMC and RH constant.

The stepwise model represents the best available linear summary of the environmental drivers of burned area in this dataset, though its modest R^2 underscores the inherent limits of predicting fire extent from weather and fuel moisture alone.

8. Model Validation

8.1 Train/Test Split

The dataset was partitioned into a training set (80%, N = 416) and a test set (20%, N = 101) using a stratified random split via the caret package (set.seed = 202031137). The full model was then re-estimated on the training data to assess whether its performance generalises to unseen observations.

8.2 Training Model Performance

The model fitted on the training data yields results broadly consistent with the full-sample estimation. The Adjusted R^2 on the training set is 0.011 — marginally higher than the full-sample value of 0.004, reflecting the typical overfitting tendency of in-sample metrics. Wind speed again emerges as the sole statistically significant predictor ($\beta = 0.083$, $t = 2.095$, $p = 0.037$). The ISI coefficient approaches significance at the 10% level ($\beta = -0.030$, $t = -1.710$, $p = 0.088$), suggesting that in the training partition, higher Initial Spread Index is negatively associated with burned area — a counter-intuitive finding that may reflect sample variation or the collinearity between ISI and wind speed.

8.3 Out-of-Sample Predictive Performance

Predictions were generated for the 101 held-out test observations and evaluated against the actual log_area values. The resulting error metrics are presented in Table 6. The Root Mean Squared Error (RMSE) of 1.71 and Mean Absolute Error (MAE) of 1.30 are expressed on the log(area + 1) scale. To contextualise these values: an MAE of 1.30 on the log scale corresponds to prediction errors of approximately $e^{1.30} \approx 3.7$ ha on the original hectare scale, which represents a substantial degree of imprecision. However, this is not surprising given the weak explanatory power established in the training phase.

Metric	Value (log scale)	Interpretation
RMSE	1.71	Average squared-root prediction error on log(area+1) scale
MAE	1.30	Average absolute prediction error on log(area+1) scale

Table 6. Out-of-sample predictive performance metrics on the test set (N = 101).

The RMSE exceeding the MAE by 0.41 confirms the presence of a small number of large prediction errors — consistent with the extreme fire observations noted throughout this analysis. The validation results collectively indicate that the current set of environmental predictors provides limited out-of-sample forecast accuracy. This finding is well-aligned with the broader wildfire literature, which reports that burned area is among the most challenging fire variables to predict and that models incorporating spatial covariates (topography, vegetation type) or suppression data substantially outperform purely meteorological specifications.

9. Interpretation and Managerial Implications

9.1 Biological Interpretation of Key Coefficients

The stepwise model identifies three environmentally meaningful drivers of fire-induced land damage. Wind speed ($\beta = 0.062$) is the strongest and most consistently significant predictor across all model specifications. Wind promotes fire spread through three mechanisms: it desiccates fuels by increasing evaporation rates, it supplies additional oxygen to the combustion zone, and it physically carries burning embers ahead of the fire front, enabling rapid spread across discontinuous fuel beds. The Duff Moisture Code ($\beta = 0.00175$) is a measure of the moisture content of loosely compacted organic layers beneath the forest floor — the very material that sustains slow-burning ground fires over extended periods. Higher DMC values indicate drier deep fuel beds, which are associated with fires that resist suppression and spread more widely. Relative humidity ($\beta = -0.00558$) directly influences the rate at which surface fuels dry and the atmospheric moisture available to slow combustion. Its negative coefficient confirms the well-established meteorological principle that lower atmospheric humidity — particularly below 30% — creates conditions highly conducive to fire ignition and spread.

Temperature, although not retained by stepwise selection, enters the time-controls model with a positive trend ($p \approx 0.095$) and is theoretically the primary long-run driver of all three retained variables: higher temperatures reduce atmospheric humidity, accelerate drying of organic soils, and increase wind velocity through convective processes. This indirect pathway through mediating variables may explain why temperature's direct effect is attenuated once DMC and RH are already in the model.

9.2 Managerial Implications

For a biotechnology firm or ecosystem management organisation working with controlled or semi-natural environments, the findings carry several practical implications. First, wind speed monitoring should be prioritised as the primary real-time fire risk indicator. Threshold-based alert systems — for example, automated warnings when wind speed exceeds 5–6 km/h during drought conditions — could be integrated into agricultural or forest management software. Second, the DMC signal reinforces the importance of tracking long-term moisture accumulation. Management practices such as controlled burns during winter months (when DMC is lower and fires remain manageable) can reduce the deep fuel load that enables catastrophic summer fires. Third, the temporal analysis (model_time) reveals elevated fire risk in July (month^7 coefficient of 1.354, $p = 0.054$), which aligns with the peak of Mediterranean summer drought. Resource allocation for fire prevention and suppression should account for this seasonal pattern, concentrating field presence and equipment in July and August. Fourth, the overall low R^2 values serve as a methodological caution: predictions of burned area based solely on meteorological inputs carry large uncertainty, and decision-making under fire risk should incorporate spatial information systems (GIS layers, satellite vegetation indices) alongside the meteorological covariates analysed here.

10. Conclusion

This case study applied multiple linear regression to 517 fire events from Montesinho Natural Park, Portugal, to model the environmental drivers of burned area — a proxy for ecosystem stress. After log-transforming the heavily right-skewed response variable, three candidate models were estimated and evaluated against each other using Adjusted R^2, AIC, and BIC.

The core statistical finding is that the eight environmental predictors explain only a very small fraction of variation in burned area (Adjusted $R^2 = 0.004$ for the full model). This result is not a failure of the analytical approach but rather reflects a genuine characteristic of wildfire behaviour: the magnitude of a fire is determined not only by weather and fuel conditions at the time of ignition, but also by a complex interplay of terrain, suppression resources, and stochastic ignition timing that is absent from this dataset.

Despite the weak overall fit, the stepwise regression identifies a parsimonious and interpretable three-variable model — DMC, RH, and wind — with the lowest information criteria (AIC = 1815.52, BIC = 1836.76). Wind speed is the most consistent and statistically significant individual predictor across all model specifications, followed by relative humidity and the Duff Moisture Code. Multicollinearity diagnostics confirm that all VIF values remain below 5, validating the stability of the estimated coefficients. The Breusch-Pagan test does not detect significant heteroscedasticity ($p = 0.187$), though the Shapiro-Wilk test decisively rejects residual normality ($p < 2.2 \times 10^{-16}$), indicating that caution is warranted when interpreting p-values for individual coefficients.

Out-of-sample validation on a 20% hold-out test set yields RMSE = 1.71 and MAE = 1.30 on the log scale, confirming that prediction uncertainty remains substantial. Future work should consider incorporating spatial covariates such as slope, aspect, and vegetation cover index; applying robust regression methods to mitigate the influence of extreme fire observations; and exploring non-linear machine learning models (e.g., gradient-boosted trees or random forests) that may better capture the threshold effects and interactions inherent in fire ecology. Nonetheless, the present analysis provides a rigorous statistical foundation and actionable insights for environmental monitoring, fire-risk alerting, and ecosystem management planning.

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